

The Cross Sectional Attention Model for Portfolio Construction

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Abstract

This document describes the cross sectional attention model for monthly portfolio construction. The model encodes each security from its characteristics and couples the securities across the cross section through self attention over the stock dimension, so that each security builds its own context from the others rather than receiving a single shared average, and it is fitted under a maximum Sharpe ratio regression objective. We develop the full model, present the attention mechanism formally, relate it to the pooled average it generalises, and interpret the empirical finding that on this panel the richer coupling does not improve, and in fact slightly reduces, the realised Sharpe ratio.

1 Overview and motivation

A simple way to couple the cross section is through a single pooled context, the mean of the security embeddings, which is the same vector for every security and lets each security’s score depend on the average of the month. This is the simplest possible form of cross sectional interaction. A natural generalisation is to let each security attend selectively to the others, so that its context is a weighted combination of their embeddings with weights that depend on the securities themselves. Self attention provides exactly this mechanism, and the present model adopts it in place of the mean pool while leaving the remainder of the network unchanged.

2 The data pipeline and objective

The raw inputs are a panel of monthly characteristics and a list of candidate features, and the target is the excess return of a security over the following month. A characteristic is retained if its coverage is at least ninety percent over the pre test period, which makes the selection causal and algorithmic. For each month every retained characteristic is transformed to its within month percentile rank, missing entries are set to the median before centring so that the inputs lie in the interval $[-0.5, 0.5]$, and securities missing more than one third of the characteristics are dropped. Each security is encoded to an embedding $z_i \in \mathbb{R}^d$ with $d = 128$.

The network is fitted under the maximum Sharpe ratio regression loss $(1 - \sum_i w_i r_i)^2$, which drives the realised managed return towards unity in every month and is the sample analogue of the tangency portfolio, averaged over a rolling window of at most sixty months with a fifty epoch cold start, ten epoch warm starts thereafter, three seeds, and seed averaging of the weights normalised to unit gross exposure.

3 Self attention across the cross section

The securities of a month are treated as a single sequence, and one round of multihead self attention is applied across that sequence. Let $Z \in \mathbb{R}^{n \times d}$ collect the encoded embeddings, and let $\bar{Z}$ denote

the layer normalised embeddings. For a single head the attention operation is

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_h}}\right) V, \quad Q = \bar{Z}W_Q, \quad K = \bar{Z}W_K, \quad V = \bar{Z}W_V, \quad (1)$$

where W_Q, W_K, W_V are learned projections and d_h is the per head dimension. Four heads are used and their outputs are concatenated. The attended representation is added back to the embedding through a residual connection,

$$z'_i = z_i + A_i, \quad (2)$$

and the score is read from a head applied to z'_i . The context of security i is $A_i = \sum_j \alpha_{ij} v_j$, a convex combination of the value vectors of all securities with attention weights α_{ij} that depend on the compatibility of the securities' queries and keys. We can observe that if the attention weights were forced to be uniform, the mechanism would reduce to a pooled average of the value vectors, so the attention model contains the simple pooled coupling as a special case and adds the capacity to weight the context selectively.

4 Interpretation of the empirical result

On the evaluation window the model attains a Sharpe ratio of 1.940 in the volatility scaled form and 1.935 in the unscaled form. An otherwise identical model that replaces the attention with the plain pooled average attains 2.484 on this panel, so the attention model underperforms despite strictly generalising the pooled coupling, and the shortfall therefore cannot be attributed to a lack of representational capacity.

We can conclude that the additional flexibility is not rewarded here and is, on net, harmful. Two considerations are consistent with this. First, the cross section in a month is large relative to the sixty month training window, so the attention parameters that govern which securities inform which are estimated from limited data and are prone to overfitting. Second, the evidence on this panel indicates that what matters is that a security's score depends on the composition of its cross section at all, not the precise form of that dependence, so a simple average already captures most of the available benefit and the selective weighting mainly adds estimation error. Within the limitations of the present analysis, we treat the deviation of a few slices of the ranking as noise and conclude that the pooled context is preferable to attention on this panel.

5 Concluding remarks

The cross sectional attention model is a clean test of whether a richer coupling improves on the pooled average, and on this panel it does not. The finding supports the interpretation that the pooled context is not merely a cheap approximation to attention but is close to the appropriate level of complexity for the cross sectional coupling given the amount of data available. The model is retained principally for the strength of this negative result rather than for its standalone performance.